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OPTION 2 · RECONSTRUCTION

On-device Phone Splatting

Scaniverse — the phone captures and trains the splat itself.

Backup

In one line: the **Scaniverse** app captures and trains the splat **entirely on your phone** — no internet, no PC — then you export it. Fast when it works, but on-device training is fragile.



Everything up to export runs on the phone itself (pink). Great when it converges; brittle when the scene is hard.

How it works, step by step

1. Open **Scaniverse** → new scan → choose **Splat / Gaussian Splatting** mode.
2. Walk a **slow overlapping loop**, 1–2 m from surfaces, covering walls, corners, floor and objects at several heights (30–60s).
3. The phone **processes & trains** the splat (a few minutes). **Wait until it looks like a real room inside the app.**
4. **Export** → **Gaussian Splat** → **`.ply`** (or `.spz`).
5. SuperSplat to tidy, then our web viewer to host & roam.

Trade-offs

✓ Strengths

- Completely free & offline — no account
- Very fast when the scene is good
- Photoreal Gaussian splat output
- Private — nothing leaves the phone

✗ Weaknesses

- On-device training fails on hard scenes
- Blank walls / glass / low light → empty splat
- Easy to export before training finishes
- Limited by the phone's GPU

Tool: Scaniverse | Input: live capture | Output: .ply / .spz | Your compute: the phone | Cost: free

WHAT WE ACTUALLY DID

We tried **3 times**. Scan #1 exported a **mesh** (wrong type). Scans #2 & #3 were correct splat format but **empty**: 514,690 of 524,940 points sat on the origin with the rest on a seed sphere — i.e. training never converged (likely a feature-poor scene or export before processing finished).

VERDICT FOR THE COMPETITION

Keep as a fast backup. Can still work with a **cluttered, well-lit, textured** scene fully processed before export — but after 3 failures, don't rely on it as plan A.